Texas
100 MW AI Campus

100 MW IT / four 25 MW phases / Central Texas

A water-conserving, liquid-cooled capacity campus in North Temple can be a credible development option only if land, a 140 MW power pathway, regulatory clearance and bankable tenant demand are proven.

CONCEPT DESIGN BASIS	LOCAL-CURRENCY FINANCIAL CASE
100 MW IT / four 25 MW phases	USD 1,362m development CAPEX
Moores Mill / Wendland study area	USD 1,456m funding incl. carry/collateral
Peak site demand 140 MW	Equity IRR 10.2% / zero terminal sale value

Investment stance

Diligence only: approval hold and base returns below hurdle

Evidence date: 13 October 2026. Land rights, power capacity, permits, suppliers, customers and financing are not secured. This package is concept/pre-feasibility work, not certified design or construction authorization.

Package contents

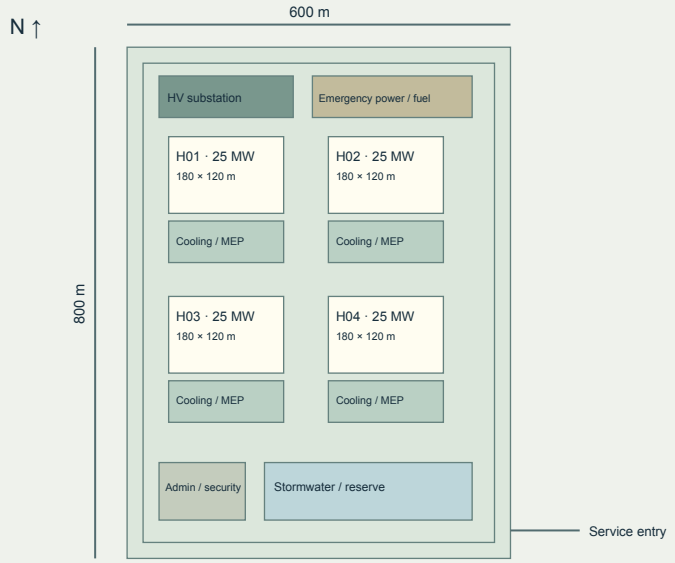
Independent site and power case; concept engineering; permitting and delivery; commercial strategy; procurement; operations and commissioning; evidence and assumptions; calculated cash flows and sensitivities.

01 / Campus test-fit

Editable vector original included in the project package. Geometry, equipment and ratings are explicit concept assumptions.

TEXAS, UNITED STATES / CONCEPT LAND ALLOCATION

Campus 600 × 800 m = 48 ha gross within a 59.5 ha historical tract



HALL FOOTPRINT

4 × 21,600 m² = 86,400 m²

BUILDING COVERAGE

18% of 48 ha campus envelope

LAYOUT STATUS

Test-fit, not a cadastral overlay

LAND CONTROL

No option, survey or access right

SITE VALIDATION

Title / easements / water / flood / geotech

POWER COMPOUND

Emergency diesel within campus yard

DESIGN GATE

11.5 ha gross outside campus envelope

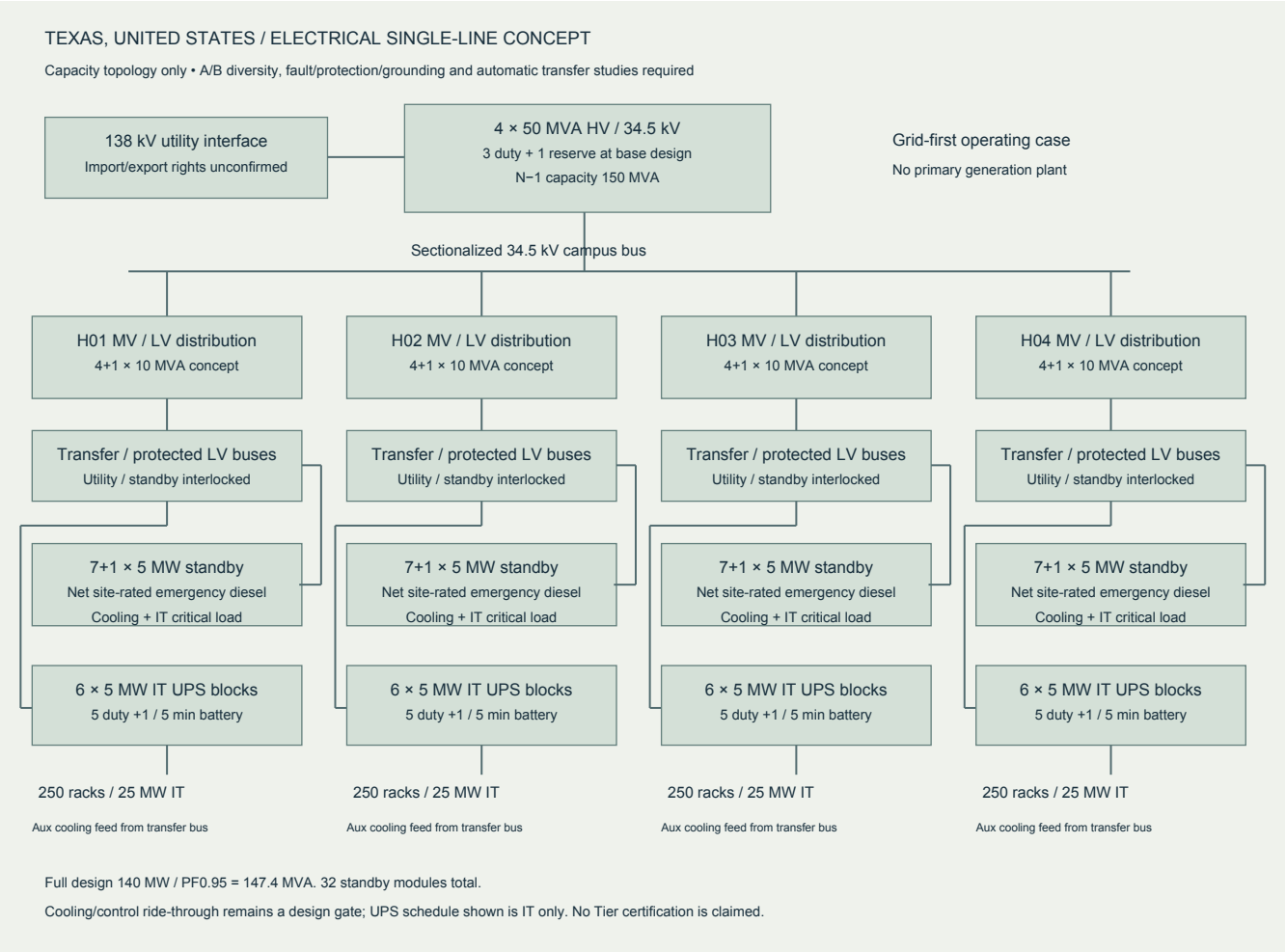
48 ha concept is illustrative. The historical 147-acre tract availability and usable boundary remain unconfirmed.

Internal roads, fire access and stormwater must be reconciled to civil engineering. Rectangles are conceptual allocations.

A rectangular land budget tests the approximate program footprint. It is not a surveyed parcel overlay; site-specific easements, environmental setbacks, drainage and road geometry require a revised test-fit.

02 / Electrical single-line

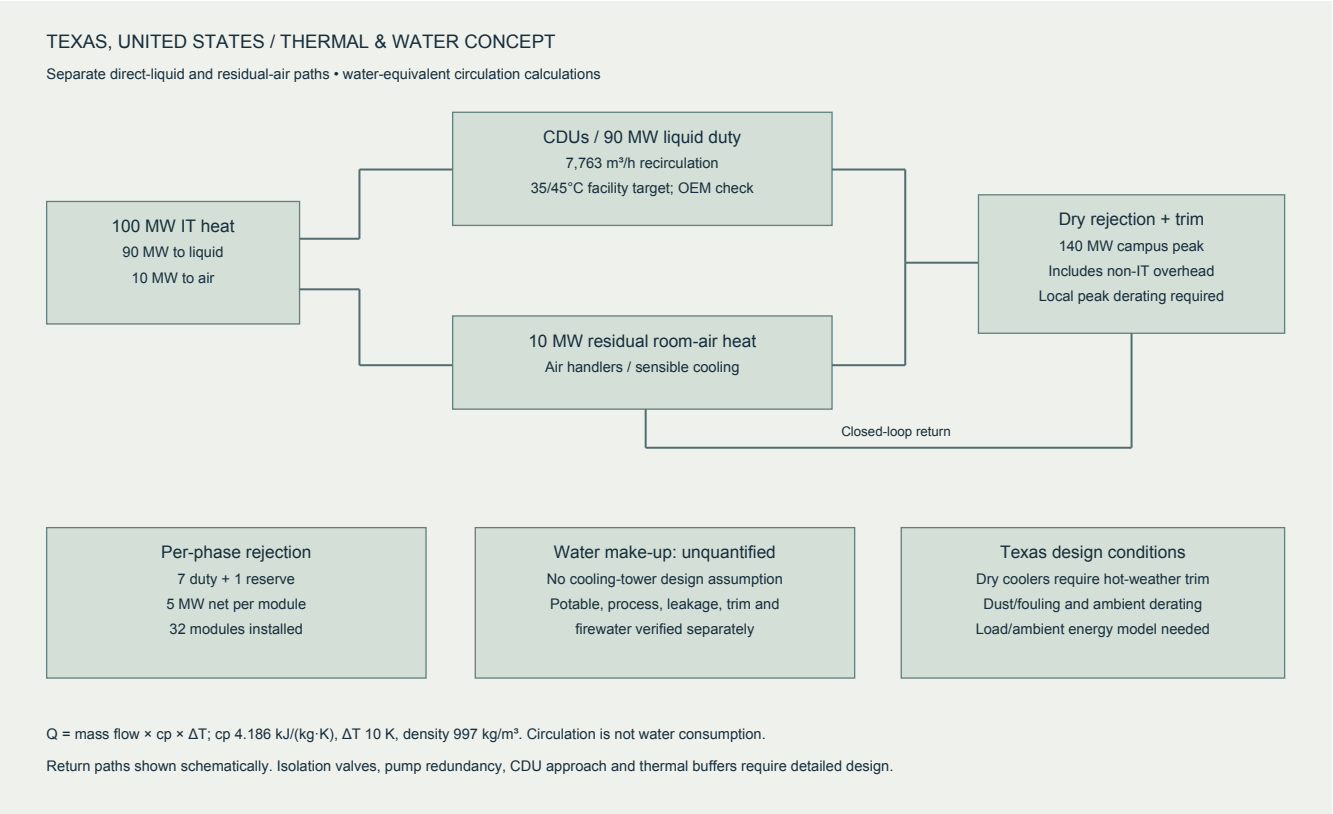
Editable vector original included in the project package. Geometry, equipment and ratings are explicit concept assumptions.



The scheme shows capacity relationships. Isolation, diverse distribution, grounding, fault levels, protection coordination, standby transfer and load acceptance require detailed engineering.

03 / Thermal concept

Editable vector original included in the project package. Geometry, equipment and ratings are explicit concept assumptions.



Annual energy efficiency and peak heat-rejection sizing are separate. Circulating flow is not water consumption. Water availability, local weather design points and manufacturer performance require confirmation.

Temple AI Campus - Texas development dossier

Concept / pre-feasibility proposal · 13 October 2026 · native USD · 100 MW IT in four 25 MW phases

This is an independently scoped facility-development project. It is not a Texas benchmark for another project. The deliverable is a candidate-site investigation, technical concept, auditable economic case and executable sequence of diligence gates. It does not constitute land control, utility approval, a permit, a construction design or an investment commitment.

Investment decision

Advance a capped diligence program; do not authorize construction or promise an energization date. Base-case equity returns also miss the proposed investment hurdle, so regulatory reopening alone would not justify FID. The North Temple industrial area is worth investigating for a water-conserving, liquid-cooled wholesale AI campus. A defensible commitment requires a real parcel, 140 MW peak power pathway, regulatory reopening, demonstrated thermal performance and a credit-supported anchor lease.

The controlling current fact is the Texas approval hold. The Governor's September 21 directive and TCEQ's current construction-stormwater notice prevent treating ordinary permitting times as a live schedule. TCEQ explicitly includes construction general-permit applications for data-center projects. The proposed calendar therefore contains an unknown period before **M0**, followed by a conditional 48-month first-phase path. October 19 is a requested government compliance update, not a promised reopening. TX05 TX06

The asset earns capacity rent and recovers metered electricity costs. It does not purchase GPUs or sell inference/training tokens. No income is assumed from energy arbitrage, generator dispatch, tax incentives or speculative land appreciation.

Candidate site and alternatives

The preferred investigation area is the **historically advertised 147 acres at the southwest corner of Moores Mill Road and Wendland Road, North Temple, Bell County**. Temple EDC's 2020 announcement identifies the address and road/water/sewer/rail context. This evidence establishes a real location to investigate; it does not establish that the land remains available. The conceptual gross study envelope is 59.5 hectares. TX01

The map marker at **31.1555256, -97.3408890** is a geographic reference on Moores Mill Road from OpenStreetMap/Nominatim. It is deliberately not presented as the tract centroid, surveyed corner or legal parcel. A diagrammed rectangular campus is an area fit study, not a survey overlay. Resolve parcel ID, owner, actual corners, setbacks, easements and developable ground before claiming this layout fits the site. TX16

Temple EDC currently publishes an industrial site-search service and identifies the regional utility context. Oncor is only a provisional power counterparty until the exact parcel's DSP/TSP confirms jurisdiction. Area telecom availability is not proof of diverse fiber. TX02 TX03

Candidate	Evidence	Decision
North Temple / Moores Mill-Wendland	Historic specific 147-acre EDC announcement; industrial access context	Preferred investigation area, subject to availability and power; no proven advantage in connection date
Five Points Business Park, Abilene	DCOA advertises over 500 acres of sites and a 50 MW substation	Retain alternate; stated substation size is below the campus's 140 MW requirement and is not evidence of spare capacity
Texas Central Park, Waco	City lists industrial development land at I-35/SH6/US84	Retain alternate; identify a real tract and utility route before scoring its power advantage

Abilene and Waco remain subject to the same statewide hold. Do not give artificial numerical rankings to unverified power schedules. If either alternative supplies a better verified connection and viable parcel, replace the North Temple selection. TX12 TX13

Parcel diligence must cover title and option terms, ALTA/NSPS survey, Phase I environmental assessment, geotechnical borings, flood and drainage modeling, wetland screening, utility and rail easements, heavy-haul access, airport/FAA constraints, neighboring residences and acoustic setbacks. No result for these items has been established.

Product and technical basis

Deliver four independently commissioned 25 MW IT blocks. The design target is concurrent maintenance of critical distribution and cooling, subject to a failure-mode study; no certification or availability percentage is claimed. Customer equipment and upstream network fabrics lie outside the landlord's capital boundary. A customer interface schedule must fix rack density, power quality, liquid temperatures/chemistry, maximum step load, rack weights, white-space scope and commissioning loads.

Capacity definitions must remain distinct:

- 100 MW IT is the campus's contracted design boundary.

- Annual PUE 1.28 is a planning estimate for energy accounting.

- Peak PUE 1.40 implies a preliminary **140 MW facility demand** at full IT design load.

- 90% committed occupancy is commercial utilization of built IT capacity.

- 85% IT load factor is average metered draw on that committed capacity.

At stabilized base assumptions, average IT draw is 76.5 MW and average facility draw is 97.92 MW. Annual purchased electricity is approximately **857,779 MWh** before further refinements to seasonal and fixed-load modeling. At USD80/MWh it costs approximately **USD68.62 million/year**, recovered as a separate electricity pass-through. It is not landlord operating profit. An actual tariff with demand charges needs an interval-load model; an all-in MWh allowance is an underwriting simplification.

The detailed electrical and thermal quantities belong to the shared engineering worksheet and accompanying concept drawings. They must be reconciled before procurement. The electrical concept uses a provisional 138 kV intake, an N+1 transformer capacity arrangement, sectional medium-voltage distribution, block UPS and emergency standby sized for design cooling auxiliaries as well as IT. Final voltage, fault duty, protection settings and grounding depend on the utility study. Dual switchboards do not establish independent utility supply.

A minimum 100 MW energy balance flows through the tenant equipment. At the assumed 90% direct-liquid capture, 90 MW goes to the liquid circuit and 10 MW remains for room-air cooling, before pumping, UPS and refrigeration heat. At a 10 K liquid temperature difference, 90 MW corresponds to roughly 2,150 kg/s water-equivalent circulation. This is **recirculated heat-transfer flow**, not daily water consumption.

Power pathway

Submit one disclosed full-campus demand/ramp. A four-phase build does not create four independent exemptions from large-load rules. The project's expected facility demand exceeds ERCOT's 75 MW large-load threshold. It has no existing queue position or demonstrated Batch Zero eligibility; the published July 2026 intake dates have passed. The first task is obtaining the applicable post-hold entry route from the interconnecting utility/ERCOT, not assuming historic serial-queue treatment. TX04

The adopted **16 TAC §25.194, effective October 8, 2026**, requires material commercial/technical disclosures and financial commitments. At 140 MW requested peak, the stated USD50,000/MW intermediate security basis yields **USD7 million**, with a **USD100,000 study fee**. Final security can be higher because it is tied to the greater of the prescribed per-MW amount and allocated upgrade costs. Confirm all tariff, credit, milestone and return/forfeiture terms with counsel and the utility. Security is separately modeled liquidity, not a second copy of construction cost. TX07, printed pages

6931-6940

Required studies include steady-state and contingency power flow, short circuit, protection coordination, dynamic behavior/ride-through, harmonics and any subsynchronous issues. Model UPS and chiller response, IT load steps, fault recovery and restoration - not simply a flat MW number. Commission agreed telemetry, metering and protection and obtain explicit authorization before each energization stage.

Choose grid supply with emergency generators in the base case. A prime-power gas alternative is a separate project requiring firm gas supply and pressure, emissions/air authorization, operability and maintenance reserves, noise controls and finance. It is not a presumed route around the state data-center permit hold. Emergency equipment must not be underwritten for routine economic dispatch without explicit environmental and contractual permission.

Cooling, weather and water

Use direct-to-chip CDUs and segregated secondary circuits, dry heat rejection and air-cooled mechanical trim, with independently maintainable pump and heat-rejection modules. Mechanical trim is essential when outdoor temperature exceeds the useful dry-cooler approach to the customer loop. An annual average temperature cannot size this system.

Use 43 deg C dry bulb as a provisional engineering screen and 46 deg C as a stress case, subject to local design-weather review and vendor curves. NWS's Waco record is a regional proxy showing both hot summers and freezes; it is not the Temple design weather file. Include hail/wind damage resistance, freeze protection, heat-trace needs and loss-of-cooling recovery. Validate annual PUE using an hourly simulation at customer-required liquid temperatures. TX09

The base design has no evaporative cooling towers. Its initial domestic/service allowance is **15 m3/day**, approximately 5,475 m3/year, excluding separately estimated first fill and emergency fire events. It is not a municipal allocation. At the modeled full-campus average IT energy, that allowance would be roughly 0.008 L/kWh; this arithmetic is a scope-limited domestic-water indicator, not a measured campus WUE. Count all actual water sources, fill, makeup, discharge and construction use before making any efficiency claim. Obtain City service, fire-flow and drought terms; no groundwater wells or new water rights are assumed. TWDB's operational survey obligations are distinct from water service authorization. TX08

Two physically independent fiber paths require traced carrier routes, separate site entrances, separated duct banks and restoration commitments. Two invoices or two carrier logos do not prove diversity. No route or latency claim is presently supported.

Permissions and implementation sequence

The permit matrix in `project.json` covers state reopening, local land/building permissions, large-load studies, air authorization, construction stormwater, municipal water/fire and conditional environmental constraints. All are unfulfilled for this proposal.

The City department is the starting point for actual parcel jurisdiction, use classification, plat/site-plan, building/MEP and certificate-of-occupancy requirements. Its exact data-center treatment and adopted code editions need confirmation. Plan drainage and access with the same surveyed site; do not assume industrial marketing establishes permitted use. TX11

An air specialist must assess the entire standby fleet and fuel storage together, with engine run hours, test concurrency, emissions factors and dispersion. Determine the required NSR/PBR/Title V route and federal engine requirements. This concept does not certify eligibility for a permit by rule. TX10

Construction stormwater coverage and SWP3 must precede relevant earthworks. The statewide hold currently affects this route; a municipal building permit would not replace it. Renewals and rule changes must be checked at the actual start, since the assumed development schedule extends beyond current permit dates. TX06

Conditional development schedule

M0 starts only after lawful regulatory restart and sponsor authority. The preceding hold is unbounded and not included below. Durations are owner planning estimates, not utility or contractor commitments.

Workstream	Months	Exit evidence
Parcel, survey, legal use and environmental screen	0-6	Conditional option and no fatal site defect
Utility intake, load models, studies and connection agreements	0-18, then works through 36	Allocated capacity, approved scope/cost/ramp and security
Tenant requirements and anchor contracting	2-18	Credit-supported minimum 50 MW lease
Design, air/water/local permits	2-18	Issuable construction package and applicable permits
HV technical freeze / order	3-12	Mature utility interface, slot and separately approved cancellation exposure
HV manufacture, factory tests and delivery	12-42	All primary units on site after 30-month order-to-site allowance
HV installation and site acceptance	42-46	Protection, telemetry and electrical installation tested
Integrated commissioning and handover	46-48	Written energization permission and tenant acceptance
FID and debt close	18	Power, permits, anchor, EPC scope and downside returns pass
Site/campus utilities	18-28	Permitted civil works and backbone ready
First 25 MW block and integrated commissioning	24-48	Written energization and tenant acceptance
Additional blocks	30-54 / 36-60 / 42-66	Cumulative 50 / 75 / 100 MW accepted
Operations mobilization and warranty	42 onward	Trained staff, procedures, spares and compliance records

Procurement is conditional on adequate design certainty. The base first-operation assumption is now **M48**, providing 30 months from the M12 HV release to M42 delivery, four months for installation/SAT and two for integrated acceptance. M42 operation is not credible on this ordering sequence. Early transformer releases at M12 create at-risk deposits and cancellation liabilities before the M18 FID and require separate sponsor approval. If the sponsor waits for M18 FID to order, shift delivery to M48 and first operation to at least M54. If preliminary utility data cannot freeze the interface by M12, move the release and downstream dates; do not shorten the lead time without supplier evidence. Phase 2-4 commitments depend on tenant demand and the utility ramp, not only the initial construction plan.

Procurement and acceptance package

No supplier has quoted these quantities or dates. Planning lead allowances are 30 months for the HV package, 24 for MV switchgear/protection, 22 for UPS/standby, 18 for liquid/trim cooling, 16 for shell/civil and 12 for fiber/controls. Obtain at least two credible bids for critical equipment, define freight/tax/escalation, cancellation exposure, factory slots, spare strategy and performance warranties. Lead-time figures in this proposal are estimates.

Each request for quotation must include peak and continuous duty, Texas summer derating, fault ratings, load-step requirements, shipping dimensions, remote-monitoring/OT security, maintenance access, test procedures and clear battery limits. Owner-furnished equipment interfaces must be assigned in the EPC contract.

Commissioning progresses through design review, factory tests, receipt checks, pre-functional testing, system functional testing and integrated failure/load testing. Prove loss of utility, failed generator/UPS module, failed cooling module, sensor faults, controls communication loss, water leakage and recovery from curtailment with simulated tenant load. Verify peak

thermal performance, power metering, protection/telemetry and fiber diversity. Tenant acceptance, approved operating procedures and complete as-builts - not merely mechanical completion - start rent.

Commercial and financial underwriting

Offer 15-year wholesale capacity leases with electricity billed separately on a defined metering/PUE basis. Target a creditworthy 50 MW anchor, then a second customer or anchor expansion for the remaining phases. No counterparties have been contacted and no demand is represented as secured.

The model horizon is 20 years from the first COD, but the proposed initial lease term is 15 years per phase. It therefore includes **60, 54, 48 and 42 months of uncontracted renewal/reletting revenue** for phases 1-4. That tail assumes unchanged rent and occupancy, no downtime and no renewal CAPEX. It is an explicit underwriting assumption, not part of the initial take-or-pay lease. A lender case must haircut or remove the tail and fund reletting costs; no present customer contract supports it.

Specify liquid temperature/chemistry and density limits, capacity ramp, acceptance conditions, utility delay relief, customer fit-out boundary, rent commencement, delivery long-stop date, curtailment terms and SLA remedies. Use a parent guarantee or acceptable bank security. A flexible training workload and a strict always-on inference SLA are different products; price them accordingly.

Base planning input	Value and basis
Facility hard cost	USD10,000/kW IT; USD1.0bn at 100 MW. Shell 1,800; electrical/UPS/standby 4,000; cooling 2,200; civil/drainage 1,000; controls/security/fiber 500; testing/commissioning 500 per kW
Land	USD12m allowance; not an owner asking price
HV connection/upgrades	USD65m off-plot utility allowance; customer-owned HV substation is in facility electrical; not a study outcome
Primary power plant	Zero; emergency standby included in facility cost
Soft costs / contingency	10% / 15% on bases documented in the shared model
Utility security	USD7m separate liquidity reserve; actual amount and release timing unconfirmed
Fixed facility OPEX	USD30m/year at stabilization: property-tax 18m, insurance 4m, staff/security 6m, administration/compliance 2m
Variable facility OPEX	USD55/kW/year on the shared model's defined capacity basis
Electricity	USD80/MWh delivered placeholder; test USD55-125/MWh
Capacity rent	USD195/kW/month; test USD150-250; no signed price evidence
Occupancy / IT load factor	90% / 85%, treated separately
Debt	60%, 8% interest, 15-year amortization; no lender offer
Equity screening rate	14% nominal; no investor commitment

With the current shared model, base CAPEX is **USD1,362.4m** before interest and collateral, and total modeled funding is approximately **USD1,456.4m** including capitalized interest and the USD7m reserve. Stabilized capacity rent is **USD210.6m/year**, EBITDA **USD175.65m/year**, and cash flow before debt service after the maintenance reserve **USD165.65m/year**. Minimum annual operating DSCR is approximately **1.60**, but equity IRR is only **10.2%**, below the 14% hurdle, and equity NPV is approximately **-USD94.9m**. The model requires approximately **USD218/kW/month** capacity rent to achieve a 14% equity NPV break-even with other assumptions unchanged. This is an estimated required rent, not a

market quote. The investment recommendation therefore remains **no FID at base inputs**. Recalculate these outputs after any change.

The financial model currently spreads common land and off-plot utility capital over M1-12, before the proposed M18 FID. This is a conservative early-cash-exposure assumption, not an authorization to spend or a representation of the gated execution plan. Rephase those cash flows to actual option, utility-deposit and construction milestones before financing; the model does not yet separately price pre-FID cancellation losses.

All values are illustrative estimates in USD. The shared calculation workbook/module is authoritative for totals, phasing, debt, DSCR, unlevered/levered cash flow and sensitivity; this narrative does not substitute a separate spreadsheet formula chain. Its documented limitations, including income tax, inflation, terminal value and reserve treatment, apply.

The landed facility unit budget includes a general procurement-tax allowance, without crediting a data-center exemption. Texas publishes eligibility rules for state sales-tax relief, but no award or continuing entitlement is assumed here. Verify property-tax districts and assessed value independently; an electricity exemption is not a property-tax exemption. TX14

Use a downside combining facility CAPEX +25%, utility upgrades +USD50m, rent -15%, occupancy 75%, PUE 1.40 and a 12-month post-restart delay. Electricity largely passes through economically, but higher bills affect competitiveness, credit exposure and tenant willingness to commit. Do not label the project financeable solely because an optimistic base DSCR exceeds 1.30. Require lender-defined downside DSCR, a sponsor-approved equity hurdle and enough equity for taxes, working capital, utility collateral and uninsured delay.

Risks, gates and immediate diligence

The most material risks are the state hold, unknown power allocation/upgrades, a historic tract that may be unavailable, anchor credit/ramp, hot-weather performance, unconfirmed water/fire/fiber, generator authorization and equipment delivery. The machine-readable risk register assigns impacts, mitigation, triggers and accountable roles. None is eliminated by a polished model.

Gate order is: **G0 lawful restart > G1 site control > G2 power/water > G3 bankable demand > G4 FID > G5 staged energization > G6 accepted tenant handover.** Some evidence gathering runs in parallel, but capital release follows the relevant gates. An existing utility queue elsewhere or industrial site's advertised substation does not prove this project has secured capacity. TX15

For the next four weeks, prepare the following requests without implying they have been sent:

1. **Regulatory counsel:** current hold interpretation, lawful diligence boundary, post-hold large-load path and collateral obligations. Stop if the proposed route relies on an unconfirmed exemption. 2. **EDC/landowner:** parcel IDs, current availability, owner authority, survey, restrictions and conditional option terms. Replace the tract if it is unavailable or fails physical fit. 3. **DSP/TSP:** 140 MW peak request, four-stage IT ramp, product/UPS/generator profile, candidate POIs and study/security process. Require actual project evidence before a delivery promise. 4. **Civil/City:** flood, wetland, geotechnical and airport screening, land-use classification, municipal service/fire-flow and drought terms. 5. **OEMs/carriers/tenants:** comparable budget quotations, thermal curves, physical fiber routes and credit-supported customer requirements.

A sponsor could authorize an initial **USD250,000 desktop diligence cap** as an estimate, with separate approval for surveys/fieldwork and no nonrefundable land purchase or equipment order until the corresponding gate. A complete utility study and full pre-FID development effort require a much larger, separately approved budget plus security. These are proposed spending gates; no money has been committed.

Kill or replace the project if the land cannot be controlled; 140 MW cannot be connected on viable terms; the current approval pathway has no acceptable resolution at the sponsor's six-month review; a 50 MW creditworthy anchor cannot be contracted; or combined downside fails finance and return hurdles. Maintain the full 100 MW concept while releasing each next phase only against evidence.

Evidence and artifact register

- `execution.md`: RFQ milestones, measurable acceptance matrix, O&M roles/compliance, proposed lease terms and unsent requests.

- `project.json`: source IDs, assumptions, independent model inputs, permits, monthly schedule, procurement, commercial case, risk register and gates.

- `source-register.md`: sixteen primary/provider sources, supported claims and explicit limitations.

- Official Texas Register adoption source: follow TX07 in `source-register.md`; printed pages 6931-6940 contain the adopted §25.194 text. The research repository retains the original PDF.

- Shared engineering basis/model and generated drawings: quantitative electrical/cooling/layout/finance calculations and validation.

Sources were checked on 13 October 2026. Recheck the regulatory hold, rule/tariff implementation, owner status and agency guidance before any real action. Budgets, equipment schedules, rents and unconfirmed service capacities are labeled assumptions rather than external commitments.

Texas execution package

Temple AI Campus · 100 MW IT · 13 October 2026 · concept / pre-feasibility · USD

This appendix turns the proposal into work packages, proposed contract terms and acceptance evidence. Every date, staffing figure and target below is a planning assumption. It is not an approved operating procedure or purchase authorization. All request templates are **unsent**.

Delivery logic and capital releases

M0 is a future authorized restart after regulatory clearance. The current hold adds an unknown period before this clock. Base first acceptance is **M48**, with cumulative 50/75/100 MW IT at M54/M60/M66.

The 30-month HV allowance is measured from a technically frozen order to arrival on site, including factory tests and freight. The sequence is M12 order > M42 delivery > M46 installation/site tests > M48 integrated acceptance. An order first released at M18 FID instead yields a minimum M54 first COD on the same allowances. Utility or permit delay moves the entire path.

An M12 purchase creates real pre-FID cancellation exposure. Authorize it separately only after the utility interface is mature, the land option permits the intended development, supplier terms identify the exposure and equity can absorb a cancellation. A refundable slot reservation is not treated as a full equipment order unless the supplier commits the manufacturing start and delivery date. The shared financial model uses a simplified early common-cost cash curve; replace it with accepted contract milestones before financial close.

Work package	Tender / technical freeze	Release / delivery	Installation and acceptance	Responsible / capital gate
TSP interface and off-site works	M0-12; settle voltage, peak/ramp, fault levels and protection boundary	Study/connection scope by M18; field completion target M42-46	Energization authority before M46-48 tests	Power lead; utility agreement and funded security
Four 50 MVA primary transformers and associated HV equipment	RFQ M3-9; technical freeze M12	M12 release; 30 months to M42 on-site	M42-46 SAT; no capacity accepted before tests	EPC electrical lead; separately approved pre-FID exposure
First-block MV/LV package	RFQ M9-12; interface freeze M16	M16 release; 24 months to M40	M40-46, coordinated with protection tests	Electrical lead; conditional release then FID

Work package	Tender / technical freeze	Release / delivery	Installation and acceptance	Responsible / capital gate
First-block UPS / emergency standby	RFQ M12-16; duty and net ratings frozen M18	M18 release; 22 months to M40	M40-46 SAT; M46-48 integrated test	Electrical/air leads; compliant engine specification and permitted operating envelope
First-block CDUs / dry rejection / trim	RFQ M14-18; customer temperatures frozen M20	M20 release; 18 months to M38	M38-44 assembly; M44-48 thermal/controls tests	MEP lead; customer interface, water and structural design
Civil / first hall	Survey/design M0-18	Permitted enabling works M18-28; hall package M24 onward	Weather-tight target M36; ready for MEP and commissioning	Civil/EPC lead; permits, FID, drainage and access
Diverse external fiber	Two route proposals by M18; paths fixed M28	Binding release by M30; 12 months to M42	Route tracing, loss and failover tests M42-46	Network lead; rights of way and independent path evidence
Controls, meters, commissioning agent	Controls RFQ M20-26; independent agent engaged M24	Controls release M30; 12 months to M42	FAT/SAT and script review, then M46-48 witness	Commissioning authority; owner-approved interface and test matrix
Later blocks	Repeat technical brief with lessons from first block	Order against each COD minus lead/install/test time; phase procurement is overlapping	Accepted at M54/M60/M66 only when complete	Commercial/EPC leads; each phase lease, power and capital gate

Supplier quotations must distinguish manufacture, FAT, freight, customs/tax, storage, installation, energization and warranty start. Require net performance at design conditions, cancellation schedules, drawing-review deadlines, interface responsibilities, escalation and damages caps. Do not fund supplier deposits from debt assumed available before lender conditions precedent.

Commissioning and acceptance matrix

The owner’s engineer and tenant must approve the final test specification before equipment release. Proposed numeric targets below define what a tender must answer; they are not manufacturer guarantees. Tests use approved switching, safety and load-bank plans under competent site supervision.

System / test	Proposed measurable acceptance evidence	Required input still unresolved	Witness / release
IT capacity and metering	Demonstrate 25 MW IT continuously for an initial 8-hour capacity run per block; calibrated IT and facility meters reconcile within an agreed 1% accounting tolerance	Actual tenant load-bank profile, instrument accuracy, allowable transient envelope	Independent commissioning authority + tenant; rent start only after accepted report
Peak thermal duty	Equipment schedule supports 35 MW facility heat rejection per block at 43 deg C dry bulb with one 5 MW rejection module unavailable; document 46 deg C stress-case output and any derating	Site design weather, vendor curves, exact indoor conditions, glycol properties, fan/pump power	MEP engineer; OEM certified curves plus witnessed functional evidence; annual PUE remains modeled until measured

System / test	Proposed measurable acceptance evidence	Required input still unresolved	Witness / release
CDU/liquid path	22.5 MW liquid duty per block at the agreed 35/45 deg C facility-loop basis, with one 2.5 MW CDU unavailable; balance and record flows, pressures and temperatures	Tenant secondary-loop temperature/chemistry, HX approach, pipe losses and allowed pressure	MEP + customer engineer; no final guarantee until interface is agreed
Residual room cooling	Demonstrate 2.5 MW residual IT heat per block plus relevant room losses, within tenant rack inlet limits, under one specified component fault	Exact air-cooled equipment share, air-flow paths and inlet limits	MEP/tenant; independent room cooling path required
Primary transformer resilience	At 140 MW campus peak and PF >=0.95, 3 remaining 50 MVA units retain >=147.37 MVA aggregate capacity; document that buses and protection can actually deliver it	TSP fault levels, impedance, sectionalization and common-mode studies	Electrical engineer + utility; arithmetic alone cannot release the gate
UPS N+1 and battery	Each block sustains 25 MW IT with one 5 MW module unavailable; demonstrate the specified 5-minute delivered autonomy or an accepted equivalent test	End-of-life battery sizing, temperature/discharge margin, actual voltage/frequency limits	OEM + commissioning authority; no interruption outside the approved customer envelope
Utility failure / standby	Each block has 35 MW net standby duty after one unit unavailable; integrated utility-loss and restoration tests stay inside customer electrical and thermal envelopes	Site-derated 5 MW unit ratings, start/load-step time, transfer/protection sequence and cooling/control ride-through	Utility, OEM, commissioning authority and tenant; UPS-only IT support does not prove cooling continuity
Fuel resilience	Verify tank usable volume against an initial **24-hour design-runtime target** at agreed critical load; independently test telemetry, containment and delivery access	OEM fuel curve, legally permitted storage, replenishment SLA and outage demand; runtime is not yet proven	Environmental/fire/operations leads; revise target or CAPEX before contract if infeasible
Fault / maintenance isolation	Complete a failure-mode matrix covering each electrical/cooling isolation boundary; witness representative failures with no loss outside the contractual tolerance	Agreed single-failure set, allowed maintenance windows and tenant tolerance	Independent commissioning authority; no certified availability claim
Fiber diversity	As-built route audit confirms two separate paths and entrances; demonstrate agreed service continuity under one carrier/path failure	Latency, bandwidth, packet loss/recovery limits and verified off-site shared structures	Network engineer + tenant; two carriers alone do not pass
Water / leak response	Verify separate mains/makeup metering, measured service pressure/flow and leak alarm/valve response to an approved test; record commissioning fill separately	City allocation, fire-flow design, domestic actuals and leakage response time	Civil/MEP/fire authorities; 15 m3/day is an allowance, not a pass/fail demand guarantee

System / test	Proposed measurable acceptance evidence	Required input still unresolved	Witness / release
Controls / cyber / handover	All critical points commissioned; all critical alarms delivered and acknowledged; restore from tested configuration backup; zero open critical defects at acceptance	Asset inventory, OT threat assessment, alarm classification and tenant interfaces	Controls lead + owner; signed as-builts, training, spares and operating procedures required

The integrated proving period is an initial **72-hour owner target** under agreed varying load and approved fault demonstrations. Eight-hour individual capacity runs may form part of it. Do not promise a 43 deg C ambient site test on a particular date: use certified performance mapping plus witnessed testing, or defer seasonal verification with an agreed retention and remedy. No unavailable criterion may be silently treated as passed.

Operating organization and budget basis

At full build, assume **40 direct operations FTE** as a scoping allowance: 24 rotating critical-facility FTE covering four simultaneous posts, with six FTE per post for roster/leave/training relief, 8 day-maintenance/controls positions, 4 facility management/compliance/planning positions and 4 network/service coordination positions. This is not a staffing minimum derived from law or an OEM requirement. Validate skill mix, shift calendars, absence/training relief and local labor costs before contracting. Physical security is a separately contracted 24/7 service, included in the staff/security budget allowance rather than these 40 FTE.

The existing USD6m/year staffing/security provision can provisionally allocate USD4.8m to 40 direct FTE at USD120,000 loaded annual cost each and USD1.2m to contracted security. Both are owner estimates, not market salary or vendor quotes. They may be inadequate once overtime, benefits and the actual security-post plan are priced; replace them before FID. OEM emergency support and specialized maintenance also draw on variable OPEX and the maintenance reserve. Texas's fixed OPEX includes separate property-tax, insurance and compliance/admin allowances.

Responsibility	Staffing / service assumption	Routine control	Escalation and evidence
Incident command and dispatch	One appointed shift lead; continuous critical facilities coverage	Shift handover, alarm review, permit-to-work, change freeze during incidents	Immediate critical alarm response, owner incident commander and tenant notice under agreed SLA; timed event log
Electrical distribution, UPS and standby	Electrical specialists plus contracted OEM support	OEM- and risk-based PM, battery trends, protection maintenance, thermography under approved procedures	Duty capacity loss or unresolved protection defect stops affected maintenance/acceptance; incident and repair record
Cooling and water	Mechanical technicians and OEM specialist availability	Temperature/pressure/flow trends, leak checks, water chemistry, heat-exchanger cleanliness, freeze/hail readiness	Projected temperature breach triggers approved load-management escalation; environmental release follows response plan
Network and controls	Service coordination plus managed network/OT service	Route health, authenticated changes, configuration backup/restore, access reviews	Carrier failure to diverse path; OT incident isolates per approved plan and alerts security/tenant
Safety, environment and permits	Accountable compliance lead and specialist contractors	Permit conditions register, generator run-hour/fuel records, water reports, SWP3 during works, tank/fire inspections	Any permit limit breach escalates immediately to management; legal notification deadlines recorded by counsel

Responsibility	Staffing / service assumption	Routine control	Escalation and evidence
Commercial / asset management	Facility manager + off-site commercial/finance support	Meter-based invoices, SLA calculation, rent/credit monitoring, insurance and tax calendar	Credit-support shortfall, claims or DSCR covenant trigger to CFO and lender under agreed documents

Before each phase opens, build a computerized maintenance system containing OEM intervals, criticality, work methods, trained personnel and spare links. No generic interval in this appendix overrides the approved OEM/permit requirement. Initial on-site spares should cover sensors, control modules, approved fuses/breakers, pump seals/motors, valves/actuators, compatible UPS modules and critical fan drives. Transformer and generator major spares need a priced pooled or OEM support arrangement; they are not assumed on a shelf.

Maintain tested emergency procedures for utility loss, failed generator, cooling interruption, leak, fire, severe weather, fuel disruption, security/OT incident and evacuation. Rehearse non-disruptive scenarios quarterly as an owner planning target and revalidate after topology changes; disruptive exercises require approved windows. Annual management review reconciles actual water/energy/PUE, engine hours/emissions, PM backlog, insurance, asset condition and maintenance capital.

The compliance register must identify each permit or agreement, responsible person, record retention, reporting period, legal deadline, evidence file and escalation contact. TCEQ engine limits, TWDB reporting, City water/fire conditions, utility/ERCOT metering/protection and local occupancy requirements are project-specific. No permit numbers or reporting dates are fabricated here.

Proposed anchor lease heads of terms

These are discussion positions, not an offer, signed contract or legal opinion. Counsel and lenders must settle enforceable terms.

Topic	Proposed position / unresolved commercial choice
Premises and capacity	Dedicated suites totaling a 50 MW anchor across two accepted 25 MW blocks; remaining capacity subject to paid, milestone-limited options
Initial term	15 years separately from acceptance of each block; renewal rights and pricing negotiated separately
Base rent	USD195 per committed IT kW/month starting after objective acceptance; illustrative price, not a quote
Electricity	Separately metered cost pass-through with documented PUE/allocation method, losses, taxes, demand charges, curtailment treatment and audit rights; no landlord energy margin assumed
Credit	Parent guarantee or acceptable bank LC sized by independent credit review; replenish on use/downgrade; no unsecured AI venture assumption
Tenant boundary	Tenant owns IT/network assets and provides density, inlet/liquid-temperature, heat split, loading and load-step requirements before design freeze
Availability	Define electrical/thermal service envelope, maintenance windows and agreed curtailment; credits and caps priced after failure-mode analysis; no unproven Tier/uptime promise
Delivery and conditions	Conditional dates tied to regulatory restart, permits, utility ramp and customer design; long-stop, termination costs and extension rights explicitly negotiated
Capacity ramp	Binding take-or-pay ramp after accepted delivery; tenant-caused delay/late fit-out treatment and landlord commissioning access defined
Construction changes	Written change control allocates customer redesign, scope changes, fuel/air requirements and added redundancy to cost and schedule

Topic	Proposed position / unresolved commercial choice
Insurance / liability	Agree insured risks, deductibles, casualty/restoration, consequential-damage treatment, limitation caps, force majeure and lender step-in
End of term	Reinstatement, tenant equipment removal, environmental responsibilities, renewal notice and residual works; no automatic renewal revenue represented

The model assumes later phases reach 90% occupancy immediately and continues rent after each initial lease. The resulting uncontracted tail is 60/54/48/42 months across phases 1-4. Underwriting must explicitly evaluate renewal downtime, new fit-out costs and rent reset; the current model does not price those items.

Unsent diligence and RFQ templates

Landowner / EDC request - draft, unsent

> We are assessing a conditional 100 MW IT campus in the Moores Mill/Wendland study area. Please identify the currently available parcel IDs, ownership/authority, boundaries, asking terms, survey, title/easements, existing studies, permitted uses and utility correspondence. Would an option conditioned on planning, power and financing be available, and what fee, access, expiry and termination terms would apply? The historic 147-acre announcement is not being treated as current availability.

DSP/TSP technical intake - draft, unsent

> Please confirm the service territory, current lawful intake process and candidate connection points for a full-campus 140 MW peak facility demand, phased 25/50/75/100 MW IT. Annual PUE is estimated at 1.28 and peak at 1.40; expected metered load and UPS/standby dynamics will be supplied. Identify studies, disclosure/site-control requirements, security and fees, utility work scope, voltage/fault-duty assumptions, curtailment/ride-through requirements and the evidence needed before any staged energization date can be relied on. We claim no Batch Zero eligibility or existing capacity allocation.

Electrical OEM RFQ - draft, unsent

> Quote a budgetary campus package on the attached conceptual duty and interface schedule, identifying all exclusions. For primary transformation, assess four 50 MVA net-duty units and a provisional 138/34.5 kV interface; confirm ratings, fault duty, loss guarantees, dimensions, maintenance arrangements and shipping route. Break down engineering, approved drawings, manufacturing start, FAT, freight/site arrival, SAT and warranty. State earliest supported order-to-site lead, slot validity, deposit and cancellation exposure, escalation, spares and service response. No PO or delivery date is authorized by this request.

Cooling OEM RFQ - draft, unsent

> Assess a 25 MW IT block with 22.5 MW direct-liquid duty and 2.5 MW residual IT air duty, plus auxiliary losses. Evaluate the 35/45 deg C facility-loop concept and 35 MW design heat rejection at 43 deg C dry bulb, with 46 deg C stress results and one module unavailable. Provide capacity/power curves, mechanical-trim requirements, water use, winter protection, fluid properties, acoustic data, maintenance isolation, lead time and guaranteed test criteria. Identify every needed tenant interface input and any failure of the proposed N+1 sizing.

Carrier / City / tenant request - draft, unsent

> Carrier: supply two end-to-end route drawings, shared structures, entrance and diversity evidence, bandwidth/latency SLA, build costs and restoration terms. City: confirm parcel use, approvals, water/fire capacity and drought/discharge conditions against an initial 15 m3/day domestic/service allowance and separately designed fill/fire needs. Prospective anchor: provide credit support, capacity/ramp, thermal and power-quality requirements, curtailment tolerance and pricing response to the proposed rent plus transparent electricity pass-through. No interest or service availability will be represented without written evidence.

Returned evidence should be dated, attributed, scoped and retained in the project data room. Change the assumption register and model only after reconciling the reply's limits and interfaces. Escalate contradictory responses before releasing the next capital gate.

Calculated financial appendix

All amounts below are USD millions except rent, ratios and capacity. Generated from the same executable model used by the website. Pre-tax, un-escalated planning inputs; no terminal sale proceeds.

Development capital	USD million
Building shell and fit-out	180.00
Campus electrical, UPS and emergency diesel	400.00
Cooling, heat rejection and CDUs	220.00
On-plot civil works and drainage	100.00
Controls, security and campus fiber	50.00
Installation testing and commissioning	50.00
Land acquisition allowance	12.00
Off-plot grid connection and utility works	65.00
Dedicated primary generation plant	0.00
Design, development, legal and permitting	107.70
Concept contingency	177.71
Total development CAPEX	1362.40
Capitalized construction interest	87.00
Refundable utility collateral	7.00
Total funding requirement	1456.41
Construction equity	551.96

Stabilized lease revenue: 210.60m/year. EBITDA: 175.65m/year. Customer rent plus metered energy: 258.54 per committed kW/month. Equity IRR: 10.15%. Equity NPV at 14%: -94.86m. Minimum annual operating DSCR: 1.60x. Minimum annual DSCR including terminal debt balloon: 1.60x. Terminal debt balloon: 0.00m. Rent for zero equity NPV: 218.12 per kW/month.

Annual cash flows

Year	End IT MW	CAPEX	EBITDA	Debt service	Equity cash flow
1	0	97.40	0.00	0.00	-45.96
2	0	13.18	0.00	0.00	-5.27
3	0	263.54	0.00	0.00	-105.42
4	25	553.44	3.66	2.15	-220.07
5	75	368.96	73.19	43.05	-121.62
6	100	65.89	157.35	92.86	29.18
7	100	0.00	175.65	103.72	61.93
8	100	0.00	175.65	103.72	61.93
9	100	0.00	175.65	103.72	61.93
10	100	0.00	175.65	103.72	61.93
11	100	0.00	175.65	103.72	61.93
12	100	0.00	175.65	103.72	61.93
13	100	0.00	175.65	103.72	61.93
14	100	0.00	175.65	103.72	61.93
15	100	0.00	175.65	103.72	61.93
16	100	0.00	175.65	103.72	61.93
17	100	0.00	175.65	103.72	61.93
18	100	0.00	175.65	103.72	61.93
19	100	0.00	175.65	101.57	64.08
20	100	0.00	175.65	60.67	104.98
21	100	0.00	175.65	10.86	154.79
22	100	0.00	175.65	0.00	165.65
23	100	0.00	175.65	0.00	165.65
24	100	0.00	161.01	0.00	158.85

Sensitivities

Independent one-variable changes. Energy cost affects customer cost and passes through landlord revenue/expense at cost.

Variable	Low / high input	Low / high equity NPV
Base rent	156.00 / 234.00	-254.9 / 65.1
Facility construction cost	8000.00 / 12000.00	42.3 / -232.1
Committed occupancy	75.00 / 100.00	-225.1 / -8.1
Pass-through electricity cost	56.00 / 104.00	-94.9 / -94.9
Construction and commissioning shift	36.00 / 60.00	-99.2 / -90.9
Loan interest	6.00 / 10.00	-43.5 / -151.7

Model qualifications

- Concept estimates, not contractor quotes, grid rights, permit approvals or a financing offer.

- Pre-tax, constant nominal inputs: no inflation/escalation, corporate/income tax, depreciation, grants, terminal asset value or GPU purchases.

- PUE is an annual input, not an hourly weather simulation; low-load fixed power and demand ratchets are not modeled.

- Electricity is passed through at cost; changes affect tenant cost, not landlord EBITDA. Owned-plant energy price must contain variable generation costs only.

- No lease ramp: each commissioned phase immediately reaches the chosen occupancy. Schedule is conditional, not a forecast of approval.

- Returns use monthly timing; annual DSCR uses cash available after maintenance capital, including any final debt balloon.

- Pre-tax, un-escalated planning model; no depreciation, income tax, incentives or GPU assets.

- No terminal sale proceeds or land salvage; utility collateral returns at the modeled horizon.

- No lease-up ramp: occupancy applies from each phase commissioning date.

- Maintenance reserve is preliminary; plant major overhauls and tenant equipment levy require separate diligence.

Full input register

Input	Value
energyPriceMwh	80
basePue	1.28
peakPue	1.4
facilityCapexPerKw	10000
landCostM	12
gridCostM	65
powerPlantCostM	0
softCostPct	10
contingencyPct	15
fixedOpexM	30
variableOpexPerKwYear	55
leaseRateKwMonth	195
debtPct	60
interestPct	8
debtYears	15
equityDiscountPct	14
occupancyPct	90
loadFactorPct	85
firstOperationMonth	48
phaseIntervalMonths	6
constructionMonths	24
operatingYears	20
maintenanceCapexPct	1
utilityCollateralM	7
generationPhaseShares	0.25,0.25,0.25,0.25
rackKw	100
liquidCapturePct	90
coolingDeltaTC	10

Input	Value
coolantSpecificHeatKjKgK	4.186
coolantDensityKgM3	997
powerFactor	0.95
electricalLossPct	4
batteryMinutes	5

Source register

TX01 / 147-acre North Temple industrial site announcement

Temple Economic Development Corporation / published 2020-11-16 / accessed 2026-10-13

Open source

Supports: Identifies a 147-acre site at the southwest corner of Moores Mill and Wendland roads, with roads, water, sewer and rail access.

Limit: Historical land announcement; present availability, ownership, utility capacity and developable acreage not verified.

TX02 / Temple site selection

Temple Economic Development Corporation / published Undated; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: City-level industrial property search and advertised land inventory.

Limit: Promotional inventory does not prove the selected tract is still available.

TX03 / Temple infrastructure

Temple Economic Development Corporation / published Undated; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: Identifies Oncor as the principal area electricity provider and the City as municipal water/wastewater provider.

Limit: Exact parcel service territory, fiber routes, utility capacity and prices require written verification; telecom list contains legacy names.

TX04 / Large Load Integration

ERCOT / published Live guidance; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: Large-load integration documents, Batch Zero eligibility, July 2026 submission dates, verification and energization forms.

Limit: This new concept has no Batch Zero status or allocated MW; future intake timing is unresolved.

TX05 / Governor directs TCEQ permit halt

Office of the Texas Governor / published 2026-09-21 / accessed 2026-10-13

Open source

Supports: Directs TCEQ to halt permits sought by data centers pending ERCOT audit and refers to water reporting and community safeguards.

Limit: No reopening date is established. The October 19 compliance update is not a reopening commitment.

TX06 / Construction stormwater permit and data-center halt notice

Texas Commission on Environmental Quality / published Last modified 2026-10-07 / accessed 2026-10-13

Open source

Supports: Live notice explicitly includes construction general-permit applications associated with data centers in the halt; identifies construction stormwater/SWP3 requirements.

Limit: No project-specific applicability determination or authorization exists.

TX07 / Adopted large-load interconnection standard, 16 TAC 25.194

Texas Secretary of State / Public Utility Commission of Texas / published 2026-10-02; effective 2026-10-08 / accessed 2026-10-13

Open source

Supports: Adopted text at printed pages 6931-6940 covers 75 MW+ loads; $100,000 study fee, $50,000/MW intermediate security, and final security at least the greater of that basis or allocated upgrade costs.

Limit: Counsel must apply full rule, tariff, disclosures, milestones and credit/forfeiture provisions. Securities are not a construction-price quotation.

TX08 / Digital Infrastructure Water Use Survey

Texas Water Development Board / published 2026 survey guidance / accessed 2026-10-13

Open source

Supports: Operating digital-infrastructure recipients must report prior-year water use; non-operating projects should contact the survey team about status.

Limit: Reporting is distinct from obtaining water service or water rights.

TX09 / Waco climate normals and extremes

National Weather Service, Fort Worth / published 1991-2020 normals; live table / accessed 2026-10-13

Open source

Supports: Regional proxy records hot summers, roughly 24 days/year at or above 100 F and freezing weather.

Limit: Waco is a regional proxy, not a Temple equipment-design weather file; extreme/design hours need a site engineer.

TX10 / Air permitting requirements and routes

Texas Commission on Environmental Quality / published Last modified 2026-08-28 / accessed 2026-10-13

Open source

Supports: Air NSR, permit-by-rule, Title V, public notice and modeling routes require assessment for emission sources.

Limit: Does not establish eligibility of this generator fleet for any streamlined permit.

TX11 / Temple Building Permits and Inspections

City of Temple / published Undated; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: Official municipal building-permit department endpoint.

Limit: Initial page resolved; later fetch returned 403. Specific zoning, code edition, use determination and review times are unverified.

TX12 / Five Points Business Park

Development Corporation of Abilene / published Undated; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: Advertises over 500 acres of sites and an on-site 50 MW substation with utility infrastructure.

Limit: 50 MW nameplate is not spare capacity and is below this campus peak demand; no parcel or capacity reserved.

TX13 / Waco industrial site selection

City of Waco / published Undated; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: Lists Texas Central Park and its advertised industrial land and transport connections.

Limit: No specific parcel or 140 MW electricity allocation is established.

TX14 / Texas data center tax exemptions

Texas Comptroller of Public Accounts / published Live guidance; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: Describes qualification, application and certification for state sales-tax relief; local treatment and project eligibility require assessment.

Limit: No incentive is assumed; political proposals are not enacted law or an award.

TX15 / Oncor first-quarter 2026 results

Oncor / published 2026-05-07 / accessed 2026-10-13

Open source

Supports: Reports extensive LC&I queue and collateral associated with interconnection requests.

Limit: Queue quantities are requests, not approved demand; no inference of availability at Temple.

TX16 / Moores Mill Road geographic reference

OpenStreetMap contributors / Nominatim / published Live geographic data; accessed 2026-10-13 / accessed 2026-10-13

Open source

Supports: Nominatim returned a Moores Mill Road reference point at 31.1555256, -97.3408890.

Limit: Road reference only; not the selected tract centroid, a survey, parcel boundary or verified road intersection. Coordinates obtained using public Nominatim API.

Engineering and financial methodology

Common model and engineering basis

Status: concept / pre-feasibility, 13 October 2026. This is an inspectable calculation basis for two separate development proposals. Every equipment selection, cost, financing term and schedule here is a design assumption unless a project dossier gives a supporting source. It is not a stamped design, power reservation, cost quote, Tier certification, construction authorization or underwriting opinion. Each project retains its own native currency, inputs and commercial/power risks.

Scope and accounting boundary

Each campus has **100 MW of IT capacity in four 25 MW phases**. The modeled business owns and leases powered/cooled facility capacity. It does not buy GPUs or sell cloud compute. Tenant occupancy is contracted IT kW; tenant load factor is average metered IT consumption divided by contracted IT capacity. Those quantities are independent.

The facility hard-cost input includes the building, on-plot civil works, transformers, UPS, an **independent emergency diesel system**, cooling, controls, campus fiber and commissioning. Land and off-plot utility works are separate. If `powerPlantCostM > 0`, that is a separate continuously operating primary plant, also owned by the project. It does not replace the emergency diesel equipment in this design. Removing diesel from an alternative proposal requires changing the equipment scope and facility hard-cost estimate, not just changing the narrative. Both normal plant and standby plant require separate emissions/noise/fire reviews.

For a grid-fed project, `energyPriceMwh` is a blended delivered energy allowance, including any tariff components allocated per consumed MWh. For an owner-operated generation project, it means **variable fuel, variable maintenance, emissions/carbon and any grid energy expense per delivered facility MWh**. It must not include the primary plant's capital recovery or fixed O&M: capital is already in `powerPlantCostM`, and fixed plant O&M must be in `fixedOpexM`. That boundary avoids charging for the plant twice. Uncontracted power-price risk, peak tariff ratchets, take-or-pay gas and pass-through enforceability remain material commercial questions.

Calculation and API contract

`model.mjs` is browser-compatible JavaScript ESM, with no dependencies. All monetary outputs ending in `M` are **millions of the project's native currency**. Return percentages are percentage points, not fractions. Capacity is MW IT unless specified otherwise. All exports merge `defaults`, `project.modelOverrides`, then explicit scenario `overrides`; inputs are validated without silent type coercion.

import { defaults, calculate, annualCashFlows, sensitivity, engineering } from './model.mjs'; const result = calculate(project, { occupancyPct: 85 });

Export / output	Meaning
`defaults`	Frozen numerical fallback assumptions; regional overrides are the actual project basis.
`calculate(project, overrides)`	`{currency, inputs, capex, stabilized, finance, timeline, cashFlows, warnings}`.
`capex`	`facilityM, landM, gridM, powerPlantM, hardCostM, softCostM, contingencyM, totalM, breakdown[]`. Breakdown rows contain `id, label, amountM, basis`. Excludes IDC and refundable collateral.

Export / output	Meaning
`stabilized`	`occupiedItMw, averageItMw, averageFacilityMw, annualItMwh, annualFacilityMwh, baseRentM, electricityRevenueM, energyCostM, totalRevenueM, fixedOpexM, variableOpexM, ebitdaM, maintenanceCapexM, cfadsM, customerAllInKwMonth`.
`finance`	IDC, funding, debt draw, construction equity, peak cumulative equity deficit, total equity contributions, equity/unlevered NPV and IRR, IRR status, minimum annual DSCR, equity/unlevered break-even rent, yield on construction cost, terminal value/debt balloon, collateral and total cash flows. See exact names in the source.
`timeline`	`firstOperationMonth, finalOperationMonth, lastModelMonth, operatingYears, phases[]`. Each phase has phase number, IT MW, cumulative IT MW, construction start and operation month.
`annualCashFlows(project, overrides)`	Same annual array as `calculate().cashFlows`, without computing IRRs.
`cashFlows[]`	Model year, first/last month in year, operating months, annual-equivalent average online IT MW, ending online IT MW, facility MWh, ending debt balance and DSCR. Cash columns: `capexM, collateralDepositM, collateralReturnM, debtDrawM, capitalizedInterestM, baseRentM, electricityRevenueM, energyCostM, fixedOpexM, variableOpexM, ebitdaM, maintenanceCapexM, cfadsM, interestM, principalM, balloonM, debtServiceM, unleveredCashFlowM, equityCashFlowM`.
`sensitivity(project, overrides)`	Six one-variable-at-a-time rows with `variable, label, unit, low, base, high, lowNpvM, baseNpvM, highNpvM, lowIrrPct, highIrrPct, lowCustomerCostKwMonth, highCustomerCostKwMonth`. Other inputs remain at the scenario basis.
`engineering(project, overrides)`	Design MW/MVA, PUE, rack count, electrical losses, plus `transformer, phaseTransformer, ups, standby, cooling, qualifications`. Engineering sizes to `peakPue`; operating energy uses `basePue`.

Undefined financial ratios are `null`, never fabricated as zero or Infinity. In particular no-debt DSCR is null; zero-occupancy break-even rent and tenant cost are null; IRR without a unique root found in the scan domain is null. Error messages identify the offending input. The API does not assert that valid numeric inputs are commercially or technically achievable.

Operating equations

At full commissioned campus size:

Occupied IT MW = 100 x occupancy / 100 Average IT MW = occupied IT MW x load factor / 100 Annual IT MWh = average IT MW x 8,760 Annual facility MWh = annual IT MWh x annual PUE Base rent = occupied IT MW x 1,000 x rent per kW-month x 12 Electricity revenue = energy expense = annual facility MWh x price per MWh Variable operating expense = occupied IT kW x variable expense per kW-year EBITDA = base rent + electricity revenue - electricity expense - fixed OPEX - variable OPEX Maintenance capital = (facility + primary-generation hard cost) x annual maintenance-capital rate CFADS = EBITDA - maintenance capital

Each commissioned phase immediately reaches the chosen steady occupancy; no customer ramp, bad debt, outage, SLA credit or collection lag is assumed. Fixed OPEX and maintenance capital scale with commissioned phases, including at zero occupancy. Actual sitewide staffing and insurance may be more front-loaded. At zero tenant load the model has zero metered energy: actual unloaded UPS, security, lighting and mechanical energy would worsen the result. Annual PUE is constant through occupancy/load scenarios and requires low-load and hourly validation. All years use 8,760 hours; no leap-year effect is modeled.

Pass-through changes deliberately do **not** create landlord EBITDA. Energy cost still changes the customer's all-in payment and competitive viability. `customerAllInKwMonth` divides base rent plus energy by committed kW-month, not metered kWh.

The maintenance reserve is a level annual capital allowance, not an OEM overhaul forecast or debt-funded expense. Fixed OPEX must cover operating labor, insurance, property/local taxes, security, routine facilities/plant maintenance and administration; variable OPEX must not duplicate the variable costs inside the energy input.

Construction, financing and valuation

Hard cost equals facility cost plus land, grid connection and primary generation. Soft cost is a percentage of **all hard costs including land**; contingency is a percentage of **hard plus soft costs**. This mechanical base is explicitly visible to avoid ambiguous percentage application. Soft cost is not a loan fee. Facility hard cost is allocated 18% shell, 40% electrical/UPS/standby, 22% cooling, 10% civil, 5% controls/security/fiber and 5% commissioning; these are concept allocations, not a quantity-survey estimate.

The engine runs monthly, beginning at M0. Land and off-plot utility cost, including their proportional soft cost and contingency, are spent uniformly M1-M12. Facility cost is allocated equally to four phases. Primary-generation cost uses `generationPhaseShares`, four validated **cost fractions**, summing to one; equal quarters are the fallback. These are distinct from the fraction of physical gas modules installed in each phase.

Alberta installs 6 / 9 / 13 / 16 cumulative gas modules, or incremental engine-count fractions 6/16, 3/16, 4/16 and 3/16. Its CAD360m plant allowance separately allocates **all CAD55m of shared gas lateral (CAD25m) and BESS (CAD30m) to phase 1**, because both must be available for first operation. The remaining CAD305m is allocated by incremental engine counts, giving phase plant costs **CAD169.375m / 57.1875m / 76.25m / 57.1875m**. Accordingly `generationPhaseShares` is `[0.4704861111111111, 0.15885416666666666, 0.21180555555555555, 0.15885416666666666]`. These cost shares do not alter the 6 / 9 / 13 / 16 module schedule. Engine-related balance of plant and dedicated civil/electrical works within the CAD305m still follow a proportional approximation; a quantity-based procurement estimate may require additional early common expenditure.

Facility and allocated plant costs are each spent evenly during the relevant phase's `constructionMonths`, ending one month before that phase's operation month. The full gas lateral and BESS allowance is therefore spent before first operation, but its exact supplier deposit, manufacturing milestone and site-payment timing is **not modeled**. Maintenance capital follows commissioned facility and commissioned plant cost separately. All land/grid common costs are allocated equally to the four phase loan tranches. This assumes early, at-risk common expenditure and does **not** implement the real project's conditional capital-release gates; actual land options, reservation deposits, cancellation exposure, civil sequencing and grid security requirements need a procurement-specific funding curve. The schedule sensitivity moves both phased construction spending and operation dates, while common M1-M12 costs stay fixed. It is not a sunk-cost delay scenario: deferring a negative-NPV project may make its present loss smaller, which does not establish a benefit from construction delay.

Each tranche draws `debtPct` of its construction spending; equity funds the remainder. Monthly interest is `interestPct / 12` times opening balance plus half of that month's new draw. Before a phase operates, interest is capitalized and fully debt financed. IDC is shown separately from construction CAPEX, added to financed principal, and is not also subtracted as an equity-funded cost. Therefore `debtPct` is the proportion of construction CAPEX financed, **not** the final debt-to-total-funding ratio. A lender has not committed to financing IDC.

Starting in the phase's first operation month, its loan pays level monthly principal and interest over `debtYears`. At zero interest, payments are straight-line principal. There is no interest-only operating period, refinancing or sculpting. A remaining balance at the model endpoint is explicitly repaid as a terminal balloon. Monthly identities are preserved in annual aggregation:

Closing debt = opening debt + construction draws + capitalized interest - principal - final balloon Unlevered CF = CFADS - construction CAPEX - collateral deposit + collateral refund Equity CF = unlevered CF + construction debt draws - cash interest - principal - final balloon

Utility collateral is assumed cash funded by equity at M1, locked throughout the model, and refunded at the endpoint; it is not a cost, rent or depreciable asset. This is a conservative liquidity convention, not a contractual statement about when a utility returns security. Earlier milestone release, letter-of-credit alternatives, interest on cash and financing fees are

excluded. `constructionEquityM` excludes subsequent operating losses; `maximumCumulativeEquityDeficitM` shows the largest cumulative net equity funding requirement assuming interim cash is retained, while `totalEquityContributionsM` totals negative equity cash flows without netting interim distributions. Neither includes additional lender reserves or discretionary working capital.

The horizon is **20 operating years from the first phase**, plus construction. Later phases therefore have fewer than 20 modeled operating years. Partial calendar years are retained. This deliberately assigns **zero terminal enterprise value and zero land residual value**, with no terminal sale costs or decommissioning expense. Debt is still fully settled. Expanding the horizon changes that assumption; a resale multiple is not hidden in returns.

NPV discounts each monthly cash flow using `(1 + annual hurdle)^(month/12)`. Both equity and unlevered NPV use the displayed `equityDiscountPct` as a common user-selected hurdle. The unlevered number is not an independently estimated WACC valuation. IRR is an annual effective rate calculated from monthly timing. The solver scans annual log-growth rates from -99.99% to 10,000%, bisects sign-changing roots, and reports a value only when one scanned root is found. Nonconventional cash flows can have multiple roots; the scan is not a symbolic proof against arbitrarily close/tangent roots, and `equityIrrStatus` / `unleveredIrrStatus` expose that qualification.

DSCR is annual CFADS divided by that year's cash interest plus principal, including a terminal balloon. Construction equity is not in the numerator. The minimum is across years with debt service; cash-flow insolvency is not hidden by clipping negative ratios. `minOperatingDscr` separately excludes the terminal balloon from the denominator, so normal operating coverage and terminal liquidity can be distinguished. Neither is a lender covenant definition. Yield on construction cost is stabilized EBITDA / CAPEX, before maintenance capital and IDC. Break-even base rent solves a **zero equity NPV at the chosen hurdle**; the separate unlevered break-even solves zero unlevered NPV. It is not merely an operating break-even.

Cash flows are nominal local-currency amounts with **zero escalation/inflation assumed**. Corporate/income tax, tax depreciation, sales-tax timing, depreciation shields, tax incentives/grants, loan/arrangement fees, debt-service reserve, construction revenue, receivable/payable working capital, compute hardware and revenue, terminal asset value and decommissioning costs are excluded. The outputs are pre-tax scenario results. No FX comparison is performed.

Electrical and thermal concept

Item	Calculation / sizing rule
IT	4 x 25 MW; planning density 100 kW per rack yields 250 rack positions per phase and 1,000 total. Other rack densities round positions upward per phase while capacity remains 25 MW.
Peak facility input	`100 MW x peakPue`. Annual average PUE is used only for energy. Texas input 1.40, for example, implies 140 MW at design peak, independent of contracted occupancy. This is a preliminary sizing proxy, not an equipment/weather load calculation.
Electrical overhead	Default 4 MW of IT-relative electrical losses, with the remaining peak overhead assigned to cooling/auxiliaries. Losses are included within PUE; not added a second time.
Primary substation	50 MVA units; duty count is ceiling of peak MW / power factor / 50; add one spare. A 140 MW, 0.95 PF example requires 147.37 MVA and 3 duty + 1 spare, 200 MVA installed / 150 MVA after one loss. Power factor 0.95 is an assumption.
Phase transformation	10 MVA units; duty count is ceiling of phase peak MVA / 10; add one spare per phase. A 35 MW phase at 0.95 PF has 36.84 MVA duty: 4 duty + 1 spare, 20 installed units campus-wide.
IT UPS	5 x 5 MW duty + 1 x 5 MW spare per phase. 120 MW installed, 100 MW firm IT. Five minutes at 100 MW is 8.333 MWh delivered; at assumed 90% delivery efficiency, 9.259 MWh stored. End-of-life, discharge-rate, temperature and reserve margins must be separately specified.
Standby	Net-at-site 5 MW emergency-diesel blocks; ceiling of phase peak MW / 5 plus one spare per phase. At 140 MW campus peak this is 7 duty + 1 spare per phase: 32 units / 160 MW installed. Nameplate catalog capacity cannot substitute for derated net site capacity. Fuel runtime and refueling are unquantified.
IT liquid capture	Assumed 90% of IT heat enters liquid, leaving 10 MW residual air heat at full IT load. Rack vendors must confirm that split.
Liquid loop	Preliminary facility supply/return design **35/45 deg C**, 10 K rise. 35 deg C is within the maximum supply associated with ASHRAE W40, subject to selected IT/OEM acceptance. CDU secondary temperature and approach must be verified; thermal class alone is not an equipment warranty.
Indoor liquid flow	`mass flow kg/s = liquid kW / (4.186 kJ/kg-K x 10 K)`. At 90 MW: 2,150.024 kg/s. Using assumed density 997 kg/m3 gives 7,763.376 m3/h. This is circulating clean water, not consumption or utility withdrawal.
CDU modules	2.5 MW units; at 22.5 MW liquid duty per phase, 9 duty + 1 spare; 40 units total. Installed CDU rating must be at actual liquid temperatures and flow.
Heat rejection	Conservatively size the campus rejection boundary to all peak facility power. At 140 MW, a clean-water-equivalent 10 K circuit is 3,344.481 kg/s or 12,076.363 m3/h. Some losses are actually dissipated outdoors; this is an aggregate upper-bound heat balance, not a single mandatory loop flow.
Heat-rejection blocks	Net-at-design-ambient 5 MW blocks; ceiling of phase heat duty / 5 plus one spare per phase. At 140 MW: 7 duty + 1 spare per phase, 160 MW installed. Regional dry-cooler/mechanical-trim design must deliver that rating at local peak conditions.

Outdoor freeze-protected glycol has different heat capacity, density and viscosity from the clean-water reference. The `coolantSpecificHeatKjKgK` and `coolantDensityKgM3` inputs allow alternate preliminary checks; actual fluid concentration, minimum outdoor temperature, pressure drop, pipe diameters, pump curves and heat-exchanger approach require engineering. A 10 K temperature rise is not evidence that free cooling works in summer. Residual room-air cooling needs a separate appropriately rated air-handling/chilled-water path; the 35 deg C liquid supply is not a room-air cooling set point. Texas's mechanical trim and Alberta's freeze protection have different requirements even though their conservation equations are the same.

The full 100 MW of IT electricity ultimately becomes heat; facility electrical and mechanical consumption adds heat within the boundary. **Primary gas-plant exhaust/radiator heat and fuel energy are outside data-center PUE** and require

separate plant heat rejection/emissions design. Dry cooling does not justify a zero total-water claim: potable use, leakage, flushing, fire reserve, and any optional adiabatic cooling remain unresolved. Annual WUE cannot be inferred from loop circulation.

N+1 quantities prove only installed capacity after one module outage. They do not prove concurrent maintainability, fault tolerance or certification. The engineering package still needs sectionalized buses/valves, maintenance isolation, diverse A/B distribution, rack transfer compatibility, selectivity, grounding, arc-flash/fault studies, harmonic/dynamic-load studies, generator start and load acceptance, cooling/control ride-through and thermal buffer sizing. The UPS schedule currently reserves IT ride-through; it does not silently claim the cooling system is UPS-backed. The commercial availability SLA is conditional on those studies and a commissioned failure-mode test plan.

Evidence and limits

Primary materials accessed 13 October 2026:

1. US Department of Energy, Cooling Water Efficiency Opportunities for Federal Data Centers. Supports annual PUE = facility energy / IT energy, WUE = water / IT energy, the distinction between circulating systems and evaporative consumption, and separation of direct-liquid and residual-air paths. It does **not** support this project's numerical PUE, cost, schedule or water allowance. 2. ASHRAE Handbook, Chapter 20: Data Centers and Telecommunication Facilities. Supports rack-level load characterization, liquid-cooling interfaces, dependence on workload/low-load behavior, and the need to coordinate reliability across facility systems. Its water-class discussion identifies class labels by maximum facility supply temperature. It does **not** certify the proposed rack density, redundancy or OEM compatibility. 3. ASHRAE, 2021 Thermal Guidelines liquid-cooling reference card. Identifies W40 among facility water-supply classes. The proposed 35/45 deg C operation remains an assumption requiring equipment confirmation.

Costs and financial terms are project estimates, not drawn from those thermal references. The documentation separates source-supported definitions from author-created sizing and arithmetic so the proposal can be reviewed without confusing an assumption with a vendor commitment.

Verification

Run `node --test tests/model.test.mjs`. Tests reconcile CAPEX and monthly capacity exposure, source/use cash flows, IDC/debt balance and amortization, electricity pass-through, zero-debt and zero-interest cases, zero occupancy and invalid inputs, negative and undefined IRRs, zero-NPV break-even rents, refundable collateral and terminal balloon, and independent electrical/hydraulic/thermal equations. Sensitivities check the direction of economically relevant changes. A passing test establishes calculation consistency, not external feasibility.